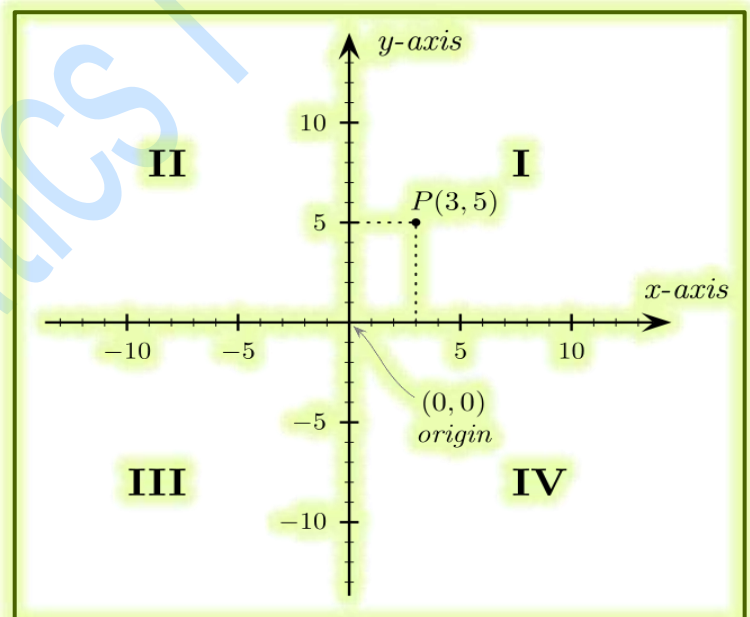


In this lecture, we will study topics about Cartesian Coordinates System: origin, x-axis, y-axis, Quadrants, I (+,+), II (-,+), III (-,-), and IV (+,-). Increment and Distance: X-Increment, Y-Increment, Distance. Straight Line Equation: Slope, parallel, orthogonal. Circle Equation: Center, Radius, Circumference, Diameter.

1.1 Cartesian Coordinates System.

The *Cartesian coordinate system*, named after the French mathematician and philosopher *René Descartes*, is a mathematical system used for locating points in a two-dimensional or three-dimensional space. It provides a way to represent geometric shapes and analyze mathematical relationships using algebraic equations.

In a two-dimensional Cartesian coordinate system, points are located using two perpendicular axes, usually labeled as the *x-axis* and *y-axis*. The point where these axes intersect is called the *origin*,



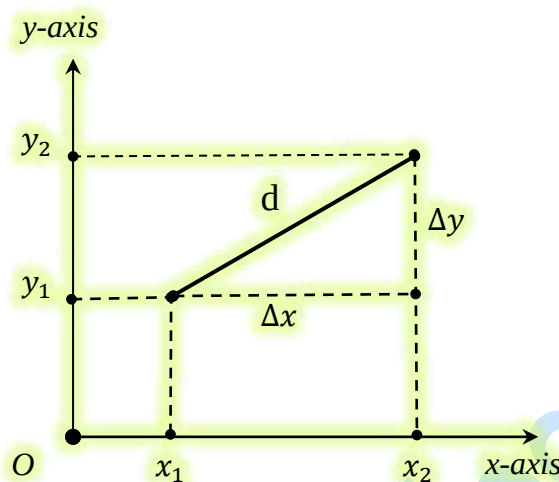
typically denoted as **(0,0)**. The *x-axis* extends horizontally to the right and left, while the *y-axis* extends vertically upwards and downwards. Each point in the plane is represented by an *ordered pair (x, y)*, where *x* is the horizontal coordinate along the x-axis, and *y* is the vertical coordinate along the y-axis.

1.2 Increment and Distance.

The change in coordinates is called increment, when a particle moves in the coordinate system from (x_1, y_1) to (x_2, y_2) , its increments are

$$\Delta x = x_2 - x_1$$

$$\Delta y = y_2 - y_1$$



And using the Pythagorean theorem, we can deduce a formula for the distance between two points (x_1, y_1) , (x_2, y_2) in the coordinate system to be,

$$\begin{aligned} d &= \sqrt{\Delta x^2 + \Delta y^2} \\ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \end{aligned}$$

Example 1.1 A particle Moves from point $(-2, 5)$ to the y -axis in such a way that $\Delta y = 3\Delta x$. Find its new coordinates and the moving distance.

Solution.

Let $(0, b)$ be the point where the particle meets the y -axis,

$$\Delta y = 3\Delta x \Rightarrow (y_2 - y_1) = 3(x_2 - x_1)$$

$$\Rightarrow (b - 5) = 3(0 - (-2))$$

$$\Rightarrow b - 5 = 6 \Rightarrow b = 11 \Rightarrow \text{the end point is } (0, 11)$$

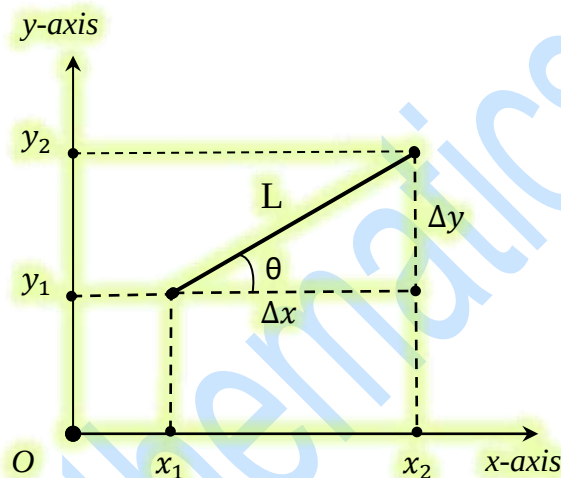
hence the distance can be calculated using the distance formula,

$$\begin{aligned} d &= \sqrt{\Delta x^2 + \Delta y^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(0 - (-2))^2 + (11 - 5)^2} \\ &= \sqrt{2^2 + 6^2} = \sqrt{4 + 36} \\ &= \sqrt{40} = 2\sqrt{10} \end{aligned}$$

1.3 Straight Line Equation.

Given a line L passing through the points from (x_1, y_1) to (x_2, y_2) , then the **slope** of L is given by,

$$\text{Slope} \equiv m = \tan(\theta) = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$



The general form of the equation of the Straight Line is given by $ax + by + c = 0$ where a, b, and c are constants. The general form can be found with one of the following methods:

- Two given points (x_1, y_1) and (x_2, y_2) and the formula:

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

- Given point (x_1, y_1) and slope m and the formula:

$$y - y_1 = m(x - x_1)$$

Example 1.2 Given the **slope-intercept** line equation $y = mx + b$, where m is the slope of the line, and b is the y-intercept. Find the slope and y-intercept of the following,

1. $-3x + y + 11 = 0$

Put the line in the slope-intercept formula, $y = 3x - 11$

From comparison the slope $m = 3$, and the y-intercept $b = -11$

2. $-2x + y - 4 = 0$

Put the line in the slope-intercept formula, $y = 2x + 4$

From comparison the slope $m = 2$, and the y-intercept $b = 4$

Example 1.3 Find the equation of the straight line passes through the point (3, 2) and make an angle $\frac{\pi}{6}$ with the *positive x-axis*.

Solution.

$$m = \tan(\theta) = \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}$$

$$y - y_1 = m(x - x_1) \Rightarrow y - 2 = \frac{1}{\sqrt{3}}(x - 3)$$

$$\Rightarrow \sqrt{3}y - 2\sqrt{3} = x - 3$$

$$\Rightarrow x - \sqrt{3}y - (2\sqrt{3} - 3) = 0$$

Example 1.4 Find the y-intercept, of a line passes through (2,2), and perpendicular to the line whose equation $y + 3x - 3 = 0$.

Solution.

Let the slope of $y + 3x - 3 = 0$ be m_2 ,

$$y = -3x + 3, \text{ Then } m_2 = -3$$

Now, the slope of the required line,

$$m_1 = \frac{-1}{m_2} = \frac{-1}{-3} = \frac{1}{3}$$

and so, the equation will be

$$(y - y_1) = m_1(x - x_1)$$

$$\rightarrow (y - 2) = \frac{1}{3}(x - 2)$$

$$\rightarrow y = \frac{1}{3}x - \frac{2}{3} + \frac{6}{3} = \frac{1}{3}x + \frac{4}{3}$$

Thus, the *y-intercept* = $\frac{4}{3}$

Example 1.5 The equations $y - x - 2 = 0$, $2y + x - 1 = 0$, and $y + 3x + 5 = 0$ are respectively the sides AB , BC , and CA of a triangle ABC . Find The equation of the altitude through B.

Solution.

The altitude through B is a line that passes through B, and is perpendicular to CA. Let CA slope be m_2 , $y = -3x - 5$, and hence $m_2 = -3$,

Now, the slope of the altitude,

$$m_1 = \frac{-1}{m_2} = \frac{-1}{-3} = \frac{1}{3}$$

Now, Find the coordinates of B, by solving the equations of the sides AB , BC simultaneously,

$$y - x = 2 \dots AB$$

$$2y + x = 1 \dots BC$$

By adding AB to BC,

$$3y = 3 \rightarrow y = 1,$$

Substitute in AB,

$$x = 1 - 2 = -1$$

Then the equation of the altitude becomes,

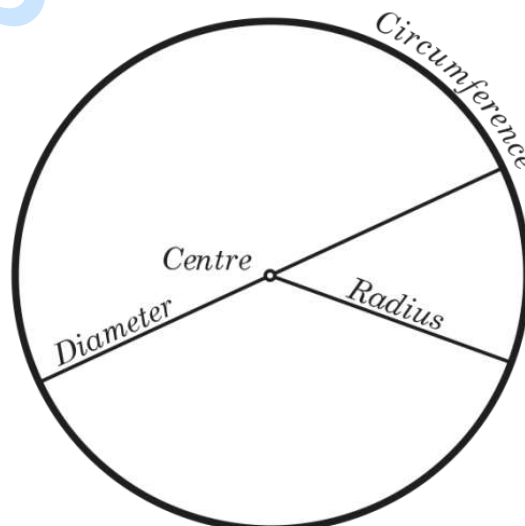
$$(y - y_1) = m_1(x - x_1)$$

$$(y - 1) = \frac{1}{3}(x + 1) \rightarrow y - \frac{1}{3}x - \frac{4}{3} = 0$$

1.4 Circle Equation.

A circle is a geometric shape that is defined as the set of all points in a plane that are equidistant from a fixed point called the center. The distance from the center to any point on the circle is called the radius. The equation of a circle in a Cartesian coordinate system with Center at $C(h, k)$ and Radius r , is given by:

$$(x - h)^2 + (y - k)^2 = r^2$$



Example 1.6 Find the equation of the circle,

- a. whose center is $C(0,0)$ and Diameter is 2 unit.

Solution.

Since the Diameter = 2 unit, then $r = \frac{1}{2} \times 2 \text{ unit} = 1 \text{ unit}$,

and the center $C(0,0)$

$\therefore x^2 + y^2 = 1$ is the equation of the circle "Unit Circle".

- b. whose center is $C(1,2)$ and Radius is 5 units.

Solution.

$r = 5 \text{ units}$, and $C(1,2)$

$$(x - 1)^2 + (y - 2)^2 = 5^2$$

$$\Rightarrow x^2 - 2x + 1 + y^2 - 4y + 4 = 25$$

$$\Rightarrow x^2 + y^2 - 2x - 4y + 1 + 4 - 25 = 0$$

$$\Rightarrow x^2 + y^2 - 2x - 4y - 20 = 0$$

is the equation of the circle

Example 1.7 Find the Center and the Radius of the circle whose equation,

$$x^2 + y^2 - 2x + 6y - 2 = 0$$

Solution.

$$\Rightarrow x^2 - 2x + 1 + y^2 + 6y + 9 = 2 + 1 + 9$$

"Completing the Square"

$$\Rightarrow (x - 1)^2 + (y + 3)^2 = 12$$

$$\therefore r = \sqrt{12}, \text{ and the center is } C(1, -3)$$

Example 1.8 Find the equation of the circle, that touches the circle,

$$x^2 + y^2 - 2x - 4y - 20 = 0$$

externally at the point (5,5), and whose radius is 5.

Solution.

The given circle is,

$$x^2 + y^2 - 2x - 4y - 20 = 0$$

$$\rightarrow x^2 - 2x + 1 + y^2 - 4y + 4 = 20 + 1 + 4$$

$$\rightarrow (x - 1)^2 + (y - 2)^2 = 25$$

Thus, the center $C_1(1,2)$,

Let the center of the circle to be found be $C_2(x_2, y_2)$, then, (5,5) is the mid-point of C_1 and C_2 as both are of the same radius.

$$\frac{x_2+1}{2} = 5$$

$$\rightarrow x_2 + 1 = 10 \rightarrow x_2 = 9$$

$$\frac{y_2+2}{2} = 5$$

$$\rightarrow y_2 + 2 = 10 \rightarrow y_2 = 8$$

Then $C_2(x_2, y_2) = (9,8)$.

Thus, the equation is

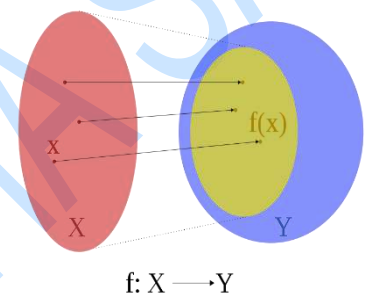
$$(x - 9)^2 + (y - 8)^2 = 25$$

Functions.

In this lecture, we will talk about Functions: Domain, Range, odd, even, Types of Functions {Polynomial, Algebraic, Transcendental (Exponential, Logarithmic, Trigonometric, Inverse Trigonometric, Hyperbolic, Inverse Hyperbolic)}.

2.1 Definition.

A function $f: X \rightarrow Y$ is a rule that assign to each element $x \in X$ one and only one $f(x) \in Y$. The set X is called the *Domain of f* and denoted by D , or it is the set of all values of x for which $y = f(x)$ is defined. The set of values assumed by f for all possible x 's is called the *Range of f* .



Example 2.1 Find the Domain and the Range for each of the following functions.

a. $y = f(x) = x^2$

$D: x \in (-\infty, \infty)$

$x = g(y) = \pm\sqrt{y}$

$R: y \geq 0$

b. $y = f(x) = \frac{1}{x-1}$

$D: x \in (-\infty, \infty) - \{1\}$

$y = \frac{1}{x-1}$

$\Rightarrow y(x-1) = 1 \Rightarrow yx - y = 1$

$\Rightarrow yx = y + 1$

$\Rightarrow x = g(y) = \frac{y+1}{y}$

$R: y \in (-\infty, \infty) - \{0\}$

c. $y = f(x) = \sqrt{1-x^2}$

$1-x^2 \geq 0 \Rightarrow$

$\Rightarrow (1-x)(1+x) \geq 0$

$\Rightarrow x \in [-1, 1]$

$x = \frac{-0 \pm \sqrt{0^2 - 4(-1)(1)}}{2(-1)}$
 $= \frac{\pm\sqrt{4}}{-2} = \frac{\pm 2}{-2} = -1, 1$

$\therefore D: x \in [-1, 1]$

d. $y = f(x) = \sqrt{x^2 - 4x + 3}$

HW

Note 1.1

- a. A function $f(x)$ is said to be odd function if $f(-x) = -f(x)$.
- b. A function $f(x)$ is said to be even function if $f(-x) = f(x)$.

Example 2.2 Test whether the following functions are odd or even.

a. $f(x) = x^3$ $f(-x) = (-x)^3 = -x^3 = -f(x)$.

∴ $f(x) = x^3$ is an odd function

b. $f(x) = \frac{1}{x^4-1}$ $f(-x) = \frac{1}{(-x)^4-1} = \frac{1}{x^4-1} = f(x)$.

∴ $f(x) = \frac{1}{x^4-1}$ is an even function

c. $f(x) = x\sqrt{x^2-1}$ HW

2.2 Types of Functions.

There is a fundamental pool of functions at the foundation of calculus and advanced calculus, these are called elementary functions. And we may describe them as follows,

a. Polynomial functions.

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

e.g. Polynomial functions.

$$f(x) = 2x^5 + 6x^3 + 3x + 1, f(x) = 3x^3 - 9x^2 + 6, f(x) = x^2 - 2x - 3.$$

b. Algebraic Function.

$$a_0(x)y^n + a_1(x)y^{n-1} + \dots + a_{n-1}(x)y + a_n(x) = 0$$

e.g. Algebraic Function.

$$y = \sqrt{1-x^2} \rightarrow y - \sqrt{1-x^2} = 0$$

And

$$y = -\sqrt{1-x^2} \rightarrow y + \sqrt{1-x^2} = 0$$

c. Transcendental Functions.

The following are some of the main types of the elementary transcendental functions:

i. Exponential Function.

$f(x) = a^x$, where $a \neq 0, 1$. If $a = e = 2.718281828 \dots$ then it is called the natural exponential function. In what follows are some of the properties of this function:

$$\begin{array}{llll} \text{a. } a^0 = 1 & \text{b. } (a^x)^y = a^{xy} & \text{c. } a^{\frac{x}{y}} = \sqrt[y]{a^x} & \text{d. } a^x a^y = a^{x+y} \\ \text{e. } a^{-x} = \frac{1}{a^x} & \text{f. } \frac{a^x}{a^y} = a^{x-y} & \text{g. } \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x} \end{array}$$

ii. Logarithmic function.

$f(x) = \log_a x$, where $a \neq 0, 1$. a “the logarithm base” can take the value of the natural exponential e and then it is called the *natural logarithmic function*, denoted by $f(x) = \log_e x = \ln x$. In what follows are some of the properties of this function:

$$\begin{array}{lll} \text{a. } \log_a a = 1 & \text{b. } \log_a 1 = 0 & \text{c. } \log_a x = \frac{\ln(x)}{\ln(a)} \\ \text{d. } \log_a(a^x) = x & \text{e. } a^{\log_a(x)} = x & \text{f. } \log_a x^p = p \log_a x \\ \text{g. } \log_a xy = \log_a x + \log_a y & & \text{h. } \log_a \frac{x}{y} = \log_a x - \log_a y \end{array}$$

Example 2.3 Simplify the following expressions.

$$\begin{array}{lll} 1. e^{\ln(2)} = 2 & 2. \ln(e^{-1.3}) = -1.3 & 3. e^{\ln(x^2+1)} = x^2 + 1 \\ 4. e^{2x+\ln(x)} = e^{2x} e^{\ln(x)} = x e^{2x} & & \\ 5. \ln\left(\frac{e^{2x}}{5}\right) = \ln(e^{2x}) - \ln(5) = 2x - \ln(5) & & \\ 6. e^{\ln(2)+3\ln(x)} & \text{HW} & \end{array}$$

Example 2.4 Solve the following equations for y.

i. $\ln(y - 1) - \ln(y) = 2x$

$$\Rightarrow \ln\left(\frac{y-1}{y}\right) = 2x \Rightarrow e^{\ln\left(\frac{y-1}{y}\right)} = e^{2x}$$

$$\Rightarrow \frac{y-1}{y} = e^{2x} \Rightarrow y-1 = ye^{2x}$$

$$\Rightarrow y - ye^{2x} = 1 \Rightarrow y(1 - e^{2x}) = 1 \Rightarrow y = \frac{1}{1-e^{2x}}$$

ii. $\ln(y - 1) = x + \ln(x)$ HW

iii. **Trigonometric Functions.**

{Degrees system $[0^\circ, 360^\circ]$, and Radian system $[0, 2\pi]$ }. The relation between the two measurement systems is given by the equation, $\frac{\text{Radian}}{\text{Degree}} = \frac{\pi}{180}$

$$\sin(x) = \frac{BC}{AB} \quad R: -1 \leq \sin(x) \leq 1$$

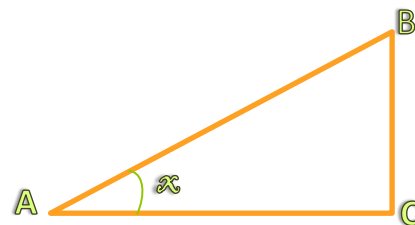
$$\cos(x) = \frac{AC}{AB} \quad R: -1 \leq \cos(x) \leq 1$$

$$\tan(x) = \frac{BC}{AC} = \frac{BC/AB}{AC/AB} = \frac{\sin(x)}{\cos(x)}$$

$$\sec(x) = \frac{AB}{AC} = \frac{1}{AC/AB} = \frac{1}{\cos(x)}$$

$$\csc(x) = \frac{AB}{BC} = \frac{1}{BC/AB} = \frac{1}{\sin(x)}$$

$$\cot(x) = \frac{AC}{BC} = \frac{AC/AB}{BC/AB} = \frac{\cos(x)}{\sin(x)} = \frac{1}{\tan(x)} \quad R: -\infty < \cot(x) < \infty$$



$$R: -\infty < \tan(x) < \infty$$

$$R: \sec(x) \leq -1 \text{ or } \sec(x) \geq 1$$

$$R: \csc(x) \leq -1 \text{ or } \csc(x) \geq 1$$

Properties of Sine.

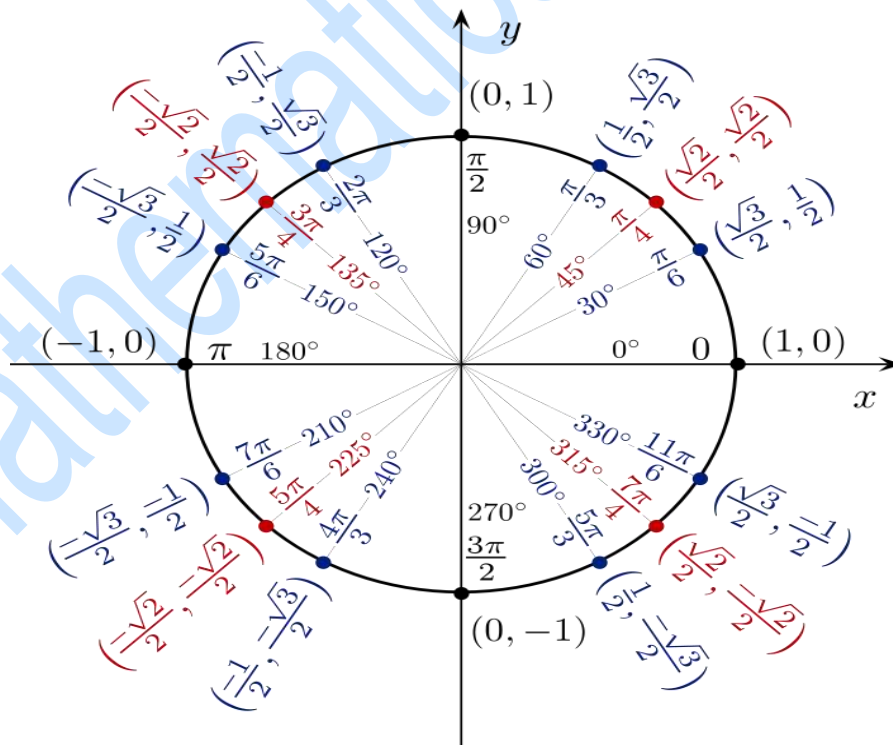
- | | | |
|---------------------------------|--|--|
| 1. $\sin(-x) = -\sin(x)$. | 2. $\sin(\frac{\pi}{2} + x) = \cos(x)$. | 3. $\sin(\frac{\pi}{2} - x) = \cos(x)$. |
| 4. $\sin(\pi + x) = -\sin(x)$. | 5. $\sin(\pi - x) = \sin(x)$. | 6. $\sin(2\pi - x) = -\sin(x)$. |

Properties of Cosine.

- | | | |
|---------------------------------|---|--|
| 1. $\cos(-x) = \cos(x)$. | 2. $\cos(\frac{\pi}{2} + x) = -\sin(x)$. | 3. $\cos(\frac{\pi}{2} - x) = \sin(x)$. |
| 4. $\cos(\pi + x) = -\cos(x)$. | 5. $\cos(\pi - x) = -\cos(x)$. | 6. $\cos(2\pi - x) = \cos(x)$. |

Identities of Trigonometric functions.

- | | | |
|--|--|-------------------------------------|
| 1. $\sin^2(x) + \cos^2(x) = 1$. | 2. $\sec^2(x) - \tan^2(x) = 1$. | 3. $\csc^2(x) - \cot^2(x) = 1$. |
| 4. $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$. | 5. $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$. | 6. $\sin(2x) = 2 \sin(x) \cos(x)$. |
| 7. $\sin(x_1 \pm x_2) = \sin(x_1) \cos(x_2) \pm \sin(x_2) \cos(x_1)$. | | |
| 8. $\cos(x_1 \pm x_2) = \cos(x_1) \cos(x_2) \mp \sin(x_1) \sin(x_2)$. | | |



Values of sine and Cosine for special angles given on the unit circle.

Example 2.5 Using the definition of trigonometric functions, prove each of the following.

a. $\sin^2(x) + \cos^2(x) = 1$.

b. $\sec^2(x) - \tan^2(x) = 1$. **HW**

c. $\csc^2(x) - \cot^2(x) = 1$. **HW**

Solution (a).

Using the definition of trigonometric functions

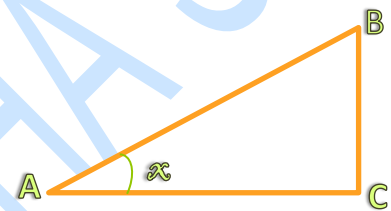
$$\sin(x) = \frac{BC}{AB} \text{ and } \cos = \frac{AC}{AB}$$

$$\sin^2(x) + \cos^2(x) = \left(\frac{BC}{AB}\right)^2 + \left(\frac{AC}{AB}\right)^2$$

$$= \frac{BC^2}{AB^2} + \frac{AC^2}{AB^2} = \frac{BC^2 + AC^2}{AB^2}$$

Using Pythagorean theorem $BC^2 + AC^2 = AB^2$, then

$$\sin^2(x) + \cos^2(x) = \frac{AB^2}{AB^2} = 1$$



iv. Inverse Trigonometric Functions.

Following we will list the Inverse Trigonometric Functions and their principal values:

$$y = \sin^{-1}(x) \quad \text{R: } 0 \leq \sin^{-1}(x) \leq 2\pi$$

$$y = \cos^{-1}(x) \quad \text{R: } 0 \leq \cos^{-1}(x) \leq 2\pi$$

$$y = \tan^{-1}(x) \quad \text{R: } \{0 \leq \tan^{-1}(x) \leq 2\pi\} - \left\{\frac{\pi}{2}, -\frac{\pi}{2}\right\}$$

$$y = \sec^{-1}(x) \quad \text{R: } 0 \leq \sec^{-1}(x) \leq 2\pi$$

$$y = \csc^{-1}(x) \quad \text{R: } 0 \leq \csc^{-1}(x) \leq 2\pi$$

$$y = \cot^{-1}(x) \quad \text{R: } \{0 \leq \cot^{-1}(x) \leq 2\pi\} - \left\{\frac{\pi}{2}, -\frac{\pi}{2}\right\}$$

Example 2.6 Solve for x.

a. $\sin^{-1}(x - 1) = \frac{\pi}{4}$

$$x - 1 = \sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

$$x = 1 + \frac{1}{\sqrt{2}}$$

b. $\tan(x + 2) = 1$

$$x + 2 = \tan^{-1}(1) = \frac{\pi}{4}$$

$$x = \frac{\pi}{4} - 2$$

c. $\cos(x^2) = -1$ HW

Example 2.7 Show each of the following.

a. $\sin^{-1}(2x\sqrt{1-x^2}) = 2\sin^{-1}(x)$

Let $x = \sin(\theta)$

$$\text{L.H.S} = \sin^{-1}\left(2\sin(\theta)\sqrt{1-\sin^2(\theta)}\right) = \sin^{-1}\left(2\sin(\theta)\sqrt{\cos^2(\theta)}\right)$$

$$= \sin^{-1}(2\sin(\theta)\cos(\theta))$$

$$= \sin^{-1}(\sin(2\theta)) = 2\theta$$

and since $\theta = \sin^{-1}(x)$ then

$$2\theta = 2\sin^{-1}(x) = \text{R.H.S}$$

b. $\sin^{-1}(-x) = -\sin^{-1}(x)$ HW

v. Hyperbolic Functions.

are defined in terms of exponential function as follows:

$$\begin{aligned}y &= \sinh(x) = \frac{e^x - e^{-x}}{2} & D: x &\in (-\infty, \infty) & R: f(x) &\in (-\infty, \infty) \\y &= \cosh(x) = \frac{e^x + e^{-x}}{2} & D: x &\in (-\infty, \infty) & R: f(x) &\in [1, \infty) \\y &= \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} & D: x &\in (-\infty, \infty) & R: f(x) &\in (-1, 1) \\y &= \operatorname{sech}(x) = \frac{2}{e^x + e^{-x}} & D: x &\in (-\infty, \infty) & R: f(x) &\in (0, 1) \\y &= \operatorname{cech}(x) = \frac{2}{e^x - e^{-x}} & D: x &\in (-\infty, \infty) - \{0\} & R: f(x) &\in (-\infty, \infty) - [-1, 1] \\y &= \operatorname{coth}(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}} & D: x &\in (-\infty, \infty) - \{0\} & R: f(x) &\in (-1, 1)\end{aligned}$$

Some identities of Hyperbolic Functions.

- a. $\sinh(-x) = -\sinh(x).$
- b. $\cosh(-x) = \cosh(x).$
- c. $\tanh(-x) = -\tanh(x).$
- d. $\sinh^2(x) = \cosh^2(x) - 1.$
- e. $\tanh^2(x) = 1 - \operatorname{sech}^2(x).$
- f. $\operatorname{coth}^2(x) = 1 + \operatorname{cech}^2(x)$
- g. $\sinh(x_1 \pm x_2) = \sinh(x_1) \cosh(x_2) \pm \sinh(x_2) \cosh(x_1).$
- h. $\cosh(x_1 \pm x_2) = \cosh(x_1) \cosh(x_2) \pm \sinh(x_1) \sinh(x_2).$

vi. Inverse Hyperbolic Functions.

<u>Function</u>	<u>Domain</u>	<u>Range</u>
$y = \sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$	$(-\infty, \infty)$	$(-\infty, \infty)$
$y = \cosh^{-1}(x) = \ln(x + \sqrt{x^2 - 1})$	$[1, \infty)$	$[0, \infty)$
$y = \tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right)$	$(-1, 1)$	$(-\infty, \infty)$
$y = \operatorname{sech}^{-1}(x) = \ln\left(\frac{1 + \sqrt{1 - x^2}}{x}\right)$	$(0, 1)$	$(-\infty, \infty)$
$y = \operatorname{cech}^{-1}(x) = \ln\left(\frac{1}{x} + \frac{\sqrt{1 - x^2}}{ x }\right)$	$(-\infty, \infty) - \{0\}$	$(-\infty, \infty) - \{0\}$
$y = \operatorname{coth}^{-1}(x) = \frac{1}{2} \ln\left(\frac{x+1}{x-1}\right)$	$(-\infty, \infty) - [-1, 1]$	$(-\infty, \infty) - \{0\}$

Example 2.8 Given the definition of the following functions,

$$\sinh(y) = \frac{e^y - e^{-y}}{2}, \quad \cosh(y) = \frac{e^y + e^{-y}}{2}, \quad \tanh(y) = \frac{e^y - e^{-y}}{e^y + e^{-y}}.$$

Derive the inverse of these functions. Do ($\operatorname{sech}(x)$, $\operatorname{cech}(x)$, $\operatorname{coth}(x)$) as **HW**.

Solution.

- a. Let $y = \sinh^{-1}(x)$, by definition of an inverse function, we want a function that satisfies the condition,

$$\begin{aligned} x &= \sinh(y) \\ &= \frac{e^y - e^{-y}}{2} = \frac{e^y - e^{-y}}{2} \cdot \frac{e^y}{e^y} = \frac{e^{2y} - 1}{2e^y} \\ 2xe^y &= e^{2y} - 1 \rightarrow e^{2y} - 2xe^y - 1 = 0 \\ &\rightarrow (e^y)^2 - 2x(e^y) - 1 = 0 \\ e^y &= \frac{2x + \sqrt{4x^2 + 4}}{2} = x + \sqrt{x^2 + 1} \\ &\rightarrow \ln(e^y) = \ln(x + \sqrt{x^2 + 1}) \\ y &= \ln(x + \sqrt{x^2 + 1}) = \sinh^{-1}(x) \end{aligned}$$

- b.** Let $y = \cosh^{-1}(x)$, by definition of an inverse function, we want a function that satisfies the condition,

$$\begin{aligned}x &= \cosh(y) \\&= \frac{e^y + e^{-y}}{2} = \frac{e^y + e^{-y}}{2} \frac{e^y}{e^y} = \frac{e^{2y} + 1}{2e^y} \\2xe^y &= e^{2y} + 1 \\&\rightarrow e^{2y} - 2xe^y + 1 = 0 \\&\rightarrow (e^y)^2 - 2x(e^y) + 1 = 0 \\e^y &= \frac{2x + \sqrt{4x^2 - 4}}{2} = x + \sqrt{x^2 - 1} \\&\rightarrow \ln(e^y) = \ln(x + \sqrt{x^2 - 1}) \\y &= \ln(x + \sqrt{x^2 - 1}) = \cosh^{-1}(x)\end{aligned}$$

- c.** Let $y = \tanh^{-1}(x)$, by definition of an inverse function, we want a function that satisfies the condition,

$$\begin{aligned}x &= \tanh(y) \\&= \frac{e^y - e^{-y}}{e^y + e^{-y}} = \frac{e^y - e^{-y}}{e^y + e^{-y}} \frac{e^y}{e^y} = \frac{e^{2y} - 1}{e^{2y} + 1} \\x(e^{2y} + 1) &= e^{2y} - 1 \\&\rightarrow x(e^{2y} + 1) - (e^{2y} - 1) = 0 \rightarrow xe^{2y} + x - e^{2y} + 1 = 0 \\&\rightarrow xe^{2y} - e^{2y} + x + 1 = 0 \rightarrow (x - 1)e^{2y} + (x + 1) = 0 \\&\rightarrow (x - 1)e^{2y} = -(x + 1) \\e^{2y} &= -\frac{(x+1)}{(x-1)} \\\ln(e^{2y}) &= \ln\left(-\frac{(x+1)}{(x-1)}\right) \rightarrow 2y = \ln\left(-\frac{(x+1)}{(x-1)}\right) \\y &= \frac{1}{2} \ln\left(-\frac{(x+1)}{(x-1)}\right) = \tanh^{-1}(x)\end{aligned}$$

Example 2.9 Show that $\sinh(\sinh^{-1}(x)) = x$.

Solution.

$$\begin{aligned}\sinh(\sinh^{-1}(x)) &= \frac{e^{\ln(x+\sqrt{x^2+1})} - e^{-\ln(x+\sqrt{x^2+1})}}{2} \\&= \frac{1}{2} \left(x + \sqrt{x^2+1} - \frac{1}{x+\sqrt{x^2+1}} \right) \\&= \frac{1}{2} \left(\frac{(x+\sqrt{x^2+1})^2 - 1}{x+\sqrt{x^2+1}} \right) \\&= \frac{1}{2} \left(\frac{x^2 + 2x\sqrt{x^2+1} + x^2 + 1 - 1}{x+\sqrt{x^2+1}} \right) \\&= \frac{1}{2} \left(\frac{2x^2 + 2x\sqrt{x^2+1}}{x+\sqrt{x^2+1}} \right) \\&= \frac{1}{2} \left(\frac{2x(x+\sqrt{x^2+1})}{x+\sqrt{x^2+1}} \right) = \frac{1}{2} (2x) = x\end{aligned}$$



Limit and Continuity of Functions.

In this lecture, we will talk about the Limit and Continuity of Functions. We will go through some topics like Theorems on Limit such as (Uniqueness of limit, Limit of constant, Obvious limit, Limit of a sum, Limit of a product, Limit of a quotient). Also, One-sided limit, Limit at infinity, Theorems on Limit at infinity, in addition to, Some special limits, Continuity.

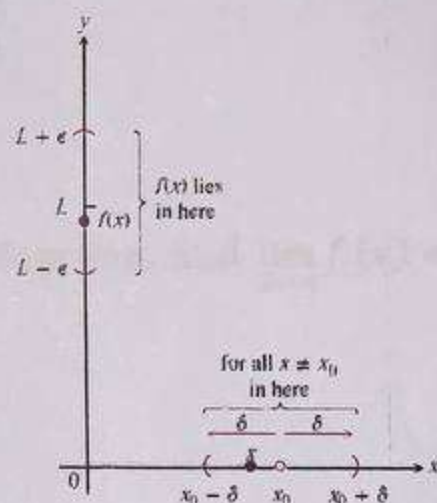
3.1 Definition.

Let $f(x)$ be defined on an open interval about x_0 , except possibly at x_0 itself. We say that the limit of $f(x)$ as x approaches x_0 is the number L , and write

$$\lim_{x \rightarrow x_0} f(x) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all x , then

$$0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \epsilon$$



3.2 Theorems on Limit (Calculation Techniques).

In what follows we shall show a group of theorems are meant to be given in this course without their proof, so we can use them as calculation techniques when we solve the limit problems,

a. Uniqueness of Limit.

If $\lim_{x \rightarrow a} f(x) = L_1$ and $\lim_{x \rightarrow a} f(x) = L_2$ then $L_1 = L_2$.

b. Limit of Constant.

If $f(x) = c$, and c is constant then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} c = c$.



c. Obvious limit.

If $f(x) = x$ then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} x = a$.

d. Limit of a sum.

If $f(x) = f_1(x) \pm f_2(x) \pm \dots \pm f_n(x)$ is a composed function, and $\lim_{x \rightarrow a} f_i(x) = L_i, i = 1, 2, \dots, n$, then

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} [f_1(x) \pm f_2(x) \pm \dots \pm f_n(x)] \\ &= \lim_{x \rightarrow a} f_1(x) \pm \lim_{x \rightarrow a} f_2(x) \pm \dots \pm \lim_{x \rightarrow a} f_n(x) \\ &= L_1 \pm L_2 \pm \dots \pm L_n.\end{aligned}$$

e. Limit of a product.

If $f(x) = f_1(x) \times f_2(x) \times \dots \times f_n(x)$ is a composed function, and $\lim_{x \rightarrow a} f_i(x) = L_i, i = 1, 2, \dots, n$, then

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} [f_1(x) \times f_2(x) \times \dots \times f_n(x)] \\ &= \lim_{x \rightarrow a} f_1(x) \times \lim_{x \rightarrow a} f_2(x) \times \dots \times \lim_{x \rightarrow a} f_n(x) \\ &= L_1 \times L_2 \times \dots \times L_n.\end{aligned}$$

f. Limit of a quotient.

If $f(x) = \frac{f_1(x)}{f_2(x)}$ is a composed function, and $\lim_{x \rightarrow a} f_1(x) = L_1$, and $\lim_{x \rightarrow a} f_2(x) = L_2 \neq 0$, then

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} \left[\frac{f_1(x)}{f_2(x)} \right] \\ &= \frac{\lim_{x \rightarrow a} f_1(x)}{\lim_{x \rightarrow a} f_2(x)} = \frac{L_1}{L_2},\end{aligned}$$



Example 3.1 Evaluate the following limits.

a. $\lim_{x \rightarrow 2} \frac{x^2 - 3x + 2}{x - 2} =$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x-1)}{x-2} = \lim_{x \rightarrow 2} (x-1)$$
$$= 2 - 1 = 1$$

b. $\lim_{x \rightarrow -1} \frac{\sqrt{2+x}-1}{x+1} =$

$$= \lim_{x \rightarrow -1} \frac{\sqrt{2+x}-1}{x+1} \times \frac{\sqrt{2+x}+1}{\sqrt{2+x}+1} = \lim_{x \rightarrow -1} \frac{(2+x)-1}{(x+1)(\sqrt{2+x}+1)}$$
$$= \lim_{x \rightarrow -1} \frac{(x+1)}{(x+1)(\sqrt{2+x}+1)} = \lim_{x \rightarrow -1} \frac{1}{\sqrt{2+x}+1}$$
$$= \frac{1}{\sqrt{2-1}+1} = \frac{1}{1+1} = \frac{1}{2}$$

c. $\lim_{x \rightarrow 2} \frac{2-x}{2-\sqrt{2x}} =$

$$= \lim_{x \rightarrow 2} \frac{2-x}{2-\sqrt{2x}} \times \frac{2+\sqrt{2x}}{2+\sqrt{2x}} = \lim_{x \rightarrow 2} \frac{(2-x)(2+\sqrt{2x})}{4-2x}$$
$$= \lim_{x \rightarrow 2} \frac{(2-x)(2+\sqrt{2x})}{2 \times (2-x)} = \lim_{x \rightarrow 2} \frac{2+\sqrt{2x}}{2}$$
$$= \frac{2+\sqrt{4}}{2} = \frac{2+2}{2} = \frac{4}{2} = 2$$

d. $\lim_{x \rightarrow 2} \frac{x^4 - 2x^2 - 8}{x^2 - 4} =$

$$= \lim_{x \rightarrow 2} \frac{(x^2-4)(x^2+2)}{(x^2-4)} = \lim_{x \rightarrow 2} (x^2 + 2)$$
$$= 4 + 2 = 6$$

e. $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} =$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{(x-1)} = \lim_{x \rightarrow 1} (x^2 + x + 1)$$
$$= 1 + 1 + 1 = 3$$

f. $\lim_{x \rightarrow 0} \frac{\sqrt{x+h}-\sqrt{h}}{x}$ H.W.



3.3 Theorems on Limit at infinity (Calculation Techniques).

In what follows a group of theorems will be shown without their proof, so we can use them as calculation techniques, when evaluating the limit problems:

- a. If $f(x) = c$, and c is constant then,

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} c = c$$

- b. If $f(x) = \frac{1}{x}$ then,

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{1}{x} = 0$$

- c. If $f(x) = \frac{1}{x}$ then ,

$$\lim_{x \rightarrow 0} f(x) = \infty$$

- d. If $a > 0$ then

$$\lim_{x \rightarrow 0^+} (\log_a x) = -\infty$$

$$\lim_{x \rightarrow \infty} (\log_a x) = \infty$$

- e. If $a > 0$ then,

$$\lim_{x \rightarrow -\infty} (a^x) = 0$$

- f. If $a < 0$ then,

$$\lim_{x \rightarrow -\infty} (a^x) = \infty$$

Remark 3.1

when dealing with the problem of evaluating the limit of a fractional function, it is noted that as x gets large, the top gets large and the bottom gets large also $\frac{\infty}{\infty}$, then the most commonly used technique: is dividing the numerator and denominator by the highest power of x in the denominator.



Example 3.2 Evaluate the following limits.

a. $\lim_{x \rightarrow \infty} \frac{3x+5}{x-4} =$

$$= \lim_{x \rightarrow \infty} \frac{\frac{3x+5}{x}}{\frac{x-4}{x}} = \lim_{x \rightarrow \infty} \frac{3+\frac{5}{x}}{1-\frac{4}{x}}$$

$$= \frac{3+0}{1-0} = 3$$

b. $\lim_{x \rightarrow \infty} \frac{x^2+2}{x^3+x^2-1} =$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x^2+2}{x^3}}{\frac{x^3+x^2-1}{x^3}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x} + \frac{2}{x^3}}{1 + \frac{1}{x} - \frac{1}{x^3}}$$

$$= \frac{0+0}{1+0-0} = 0$$

c. $\lim_{x \rightarrow \infty} \frac{2x^2+3x-1}{5x^2-2x+7} =$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2x^2+3x-1}{x^2}}{\frac{5x^2-2x+7}{x^2}} = \lim_{x \rightarrow \infty} \frac{2+\frac{3}{x}-\frac{1}{x^2}}{5-\frac{2}{x}+\frac{7}{x^2}}$$

$$= \frac{2+0-0}{5-0+0} = \frac{2}{5}$$

d. $\lim_{x \rightarrow \infty} \frac{2x^3+2x-1}{x^2-2x+5} =$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2x^3+2x-1}{x^3}}{\frac{x^2-2x+5}{x^3}} = \lim_{x \rightarrow \infty} \frac{2+\frac{2}{x}-\frac{1}{x^3}}{\frac{1}{x}-\frac{2}{x^2}+\frac{5}{x^3}}$$

$$= \frac{2+0-0}{0-0+0} = \frac{2}{0} = \infty$$



e. $\lim_{x \rightarrow \infty} \frac{x+2}{\sqrt{9x^2+1}} =$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x+2}{\sqrt{x^2}}}{\frac{\sqrt{9x^2+1}}{\sqrt{x^2}}} = \lim_{x \rightarrow \infty} \frac{\frac{x+2}{x}}{\sqrt{\frac{9x^2+1}{x^2}}}$$

$$= \lim_{x \rightarrow \infty} \frac{1+\frac{2}{x}}{\sqrt{9+\frac{1}{x^2}}} = \frac{1+0}{\sqrt{9+0}}$$

$$= \frac{1}{\sqrt{9}} = \frac{1}{3}$$

f. $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+3}}{3x-5} =$

$$= \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{x^2+3}}{\sqrt{x^2}}}{\frac{3x-5}{\sqrt{x^2}}} = \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{x^2+3}}{\sqrt{x^2}}}{\frac{3x-5}{x}}$$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{1+\frac{3}{x^2}}}{3-\frac{5}{x}} = \frac{\sqrt{1+0}}{3-0}$$

$$= \frac{1}{3}$$

g. $\lim_{x \rightarrow \infty} (\sqrt{x^2-3x}-x) =$

$$= \lim_{x \rightarrow \infty} (\sqrt{x^2-3x}-x) \left(\frac{\sqrt{x^2-3x}+x}{\sqrt{x^2-3x}+x} \right) = \lim_{x \rightarrow \infty} \frac{(x^2-3x)-x^2}{\sqrt{x^2-3x}+x}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{-3x}{\sqrt{x^2}}}{\frac{\sqrt{x^2-3x}+x}{\sqrt{x^2}}} = \lim_{x \rightarrow \infty} \frac{-3}{\frac{\sqrt{x^2-3x}}{\sqrt{x^2}} + \frac{x}{\sqrt{x^2}}}$$

$$= \lim_{x \rightarrow \infty} \frac{-3}{\sqrt{\frac{x^2-3x}{x^2}} + 1} = \lim_{x \rightarrow \infty} \frac{-3}{\sqrt{1-\frac{3}{x}} + 1}$$

$$= \frac{-3}{\sqrt{1-0}+1} = \frac{-3}{\sqrt{1}+1} = \frac{-3}{2}$$



3.4 Special limits.

Special limits refer to particular cases or types of limits that often appear in calculus and have distinct properties or results. These special limits help understand specific behaviors of functions in various situations. Some common special limits are,

$$a. \lim_{x \rightarrow 0} \frac{\sin mx}{kx} = \frac{m}{k}.$$

$$b. \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$$

$$c. \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a$$

$$d. \lim_{x \rightarrow 0} \frac{e^{mx} - 1}{x} = m$$

$$e. \lim_{x \rightarrow \infty} \left(1 + \frac{k}{x}\right)^{mx} = e^{mk}$$

$$f. \lim_{x \rightarrow 0} (1 + kx)^{\frac{m}{x}} = e^{mk}$$

$$g. \lim_{x \rightarrow 1} \frac{\ln x}{x-1} = 1$$

$$h. \lim_{x \rightarrow 0} \frac{\ln(x+1)}{x} = 1$$

Example 3.3 Evaluate the following limits.

$$a. \lim_{x \rightarrow 0} \frac{\tan x}{x} =$$

$$= \lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x}}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \lim_{x \rightarrow 0} \frac{1}{\cos x}$$

$$= 1 \times 1 = 1$$

$$b. \lim_{x \rightarrow 0} \frac{\sec x - 1}{x} =$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\cos x} - 1}{x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \cos x}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \cos x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \times \lim_{x \rightarrow 0} \frac{1}{\cos x}$$

$$= 0$$



c. $\lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x}} =$

$$\begin{aligned} &= \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{x} \times \frac{\sin x}{\sin x}} = \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{\sin x} \times \frac{\sin x}{x}} \\ &= \lim_{x \rightarrow 0} \left((1 + \sin x)^{\frac{1}{\sin x}} \right)^{\frac{\sin x}{x}} = \left(\lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{\sin x}} \right)^{\lim_{x \rightarrow 0} \frac{\sin x}{x}} \\ &= \left(\lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{\sin x}} \right)^1 = \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{\sin x}} \end{aligned}$$

when $x \rightarrow 0$, $\sin x \rightarrow 0$, and hence the limit becomes,

$$\begin{aligned} \lim_{x \rightarrow 0} (1 + \sin x)^{\frac{1}{\sin x}} &= \lim_{\sin x \rightarrow 0} (1 + \sin x)^{\frac{1}{\sin x}} \\ &= e \end{aligned}$$

d. $\lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} =$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{2^x - 1}{\sqrt{1+x} - 1} \cdot \frac{x}{x} = \lim_{x \rightarrow 0} \frac{2^x - 1}{x} \cdot \frac{x}{\sqrt{1+x} - 1} \\ &= \ln(2) \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x} - 1} = \ln(2) \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x} - 1} \cdot \frac{\sqrt{1+x} + 1}{\sqrt{1+x} + 1} \\ &= \ln(2) \lim_{x \rightarrow 0} \frac{x(\sqrt{1+x} + 1)}{1+x-1} = \ln(2) \lim_{x \rightarrow 0} \sqrt{1+x} + 1 \\ &= 2 \ln(2) \end{aligned}$$

3.5 Continuity.

Let $f(x)$ be defined for all values of x near $x = x_0$ as well as at $x = x_0$. The function $f(x)$ in a single variable x is said to be continuous at point x_0 if the following three conditions are met

- $f(x_0)$ is defined, so that x_0 is in the domain of $f(x)$.
- $\lim_{x \rightarrow x_0} f(x)$ exists for x_0 in the domain of $f(x)$.
- $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.



Remark 3.2

If two functions $f(x)$ and $g(x)$ are both continuous at x_0 , then

$$f(x) + g(x), f(x) - g(x), f(x) \times g(x), f(x) \div g(x)$$

provided that $g(x_0) \neq 0$, and $f(g(x))$ provided that $f(x)$ is continuous at $g(x_0)$, are all continuous functions at x_0 .

Example 3.4 Check whether the following functions are continuous at the given point.

1. $f(x) = x^2 - 1$, at $x_0 = 0$, and $x_0 = 1$.

a. $f(0) = -1$ defined.

b. $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} (x^2 - 1) = -1$ exist.

c. $\lim_{x \rightarrow 0} f(x) = f(0)$.

Then $f(x) = x^2 - 1$ is continuous at $x_0 = 0$.

2. $f(x) = \frac{1}{x-2} - 3x$, at $x_0 = 3$, and $x_0 = 2$.

a. $f(3) = \frac{1}{3-2} - 3 \times 3 = 1 - 9 = -8$ defined.

b. $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} (\frac{1}{x-2} - 3x) = -8$ exist.

c. $\lim_{x \rightarrow 3} f(x) = f(3)$.

Then $f(x) = \frac{1}{x-2} - 3x$ is continuous at $x_0 = 3$.



3. $f(x) = \frac{x+1}{x^2-4x+3}$ at $x_0 = 0$, $x_0 = 1$, and $x_0 = 3$.

a. $f(0) = \frac{0+1}{0-0+3} = \frac{1}{3}$ defined.

b. $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(\frac{x+1}{x^2-4x+3} \right) = \frac{1}{3}$ exist.

c. $\lim_{x \rightarrow 0} f(x) = f(0)$.

Then $f(x) = \frac{x+1}{x^2-4x+3}$ is continuous at $x_0 = 0$.

4. $f(x) = \frac{x+2}{\cos x}$ at $x_0 = \frac{\pi}{4}$, and $x_0 = \frac{\pi}{2}$.

a. $f\left(\frac{\pi}{4}\right) = \frac{\frac{\pi}{4}+2}{\cos\left(\frac{\pi}{4}\right)} = \frac{\frac{\pi+8}{4}}{\frac{\sqrt{2}}{2}} = \frac{\pi+8}{4} \times \frac{2}{\sqrt{2}} = \frac{\pi+8}{2\sqrt{2}}$ defined.

b. $\lim_{x \rightarrow \frac{\pi}{4}} f(x) = \lim_{x \rightarrow \frac{\pi}{4}} \left(\frac{x+2}{\cos x} \right) = \frac{\frac{\pi}{4}+2}{\cos\left(\frac{\pi}{4}\right)} = \frac{\frac{\pi+8}{4}}{\frac{\sqrt{2}}{2}} = \frac{\pi+8}{4} \times \frac{2}{\sqrt{2}} = \frac{\pi+8}{2\sqrt{2}}$ exist.

c. $\lim_{x \rightarrow \frac{\pi}{4}} f(x) = f\left(\frac{\pi}{4}\right)$.

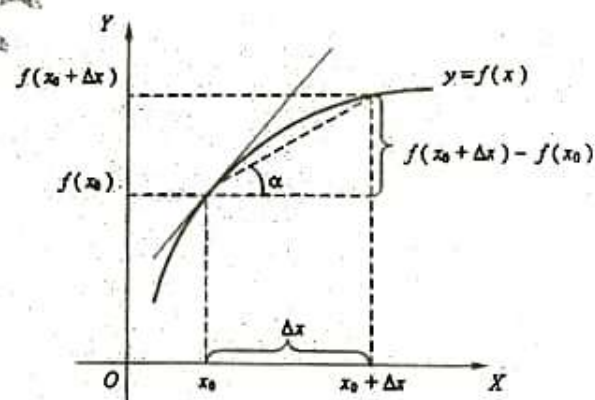
Then $f(x) = \frac{x+2}{\cos x}$ is continuous at $x_0 = \frac{\pi}{4}$.

Differentiation.

In this lecture, we will talk about the Derivative as a Function, Calculating Derivatives from the Definition. We will talk about various concepts like rules of differentiation (Constant Function, Powers of a function, Constant Multiple, Summation or Subtraction of Functions, Product of Two Functions, Quotient of Two Functions), Differentiation Formulas for several functions, differentiation techniques (Implicit differentiation, Logarithmic Differentiation, Chain Rule).

4.1 Differentiation.

Differentiation is a fundamental concept in calculus, it is the process of finding the derivative of a function $y = f(x)$, which represents the instantaneous rate of change of a function with respect to its independent variable x . Geometrically, the derivative at a specific point on the graph of a function corresponds to the slope of the tangent line at that point.



$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

There are many popular notations for writing the derivative of a function $y = f(x)$, and as shown below,

- LEIBNIZ'S notation $\frac{d}{dx} y = \frac{d}{dx} f(x)$ and higher derivatives are $\frac{d^n}{dx^n} y = \frac{d^n}{dx^n} f(x)$.
- LAGRANGE'S notation $\dot{y} = \dot{f}(x)$ and higher derivatives are $y^{(n)} = f^{(n)}(x)$.
- Newton's notation for the first derivative $\dot{y} = \dot{f}(x)$, and for the second $\ddot{y} = \ddot{f}(x)$.



Differentiation has applications to nearly all quantitative disciplines. For example, for a real-valued function of a single real variable, the derivative of a function at a point generally determines the best linear approximation to the function at that point. Furthermore, in physics the *derivative of the displacement* of a moving body with respect to time is the *velocity* of the body, and the *derivative of velocity* with respect to time is *acceleration*. In addition, Newton's second law of motion states that the *derivative of the momentum* of a body equals the *force* applied to the body. Moreover in *operations research*, derivatives determine the most efficient ways to transport materials and design factories.

4.2 Derivatives using the Definition.

In this section we will give some main rules of differentiation derived using the definition. Furthermore, we will take some examples on how to use the definition to evaluate the derivative to whatever required functions.

4.2.1 Constant Function.

If we have the function $y = f(x) = c$, where c is a constant value. Then the derivative $\dot{y} = 0$.

4.2.2 Power Function.

If we have the function $y = f(x) = x^n$, where $n > 0$. Then the derivative $\dot{y} = nx^{n-1}$.

4.2.3 Constant Multiple.

If we have the function $y = f(x) = cu(x)$, where c is a constant value and $u(x)$ is a differentiable function. Then the derivative $\dot{y} = c\dot{u}(x)$



4.2.4 Summation of Functions.

If we have a function $y = f(x) = u_1(x) \pm u_2(x) \pm \dots \pm u_n(x)$ defined as a summation or subtraction of some other differentiable functions $u_1(x), u_2(x), \dots, u_n(x)$. Then the derivative $\dot{y} = \dot{u}_1(x) \pm \dot{u}_2(x) \pm \dots \pm \dot{u}_n(x)$.

4.2.5 Product of Two Functions.

If we have the function $y = f(x) = u(x)v(x)$, where $u(x)$, and $v(x)$ are differentiable functions. Then the derivative $\dot{y} = u(x)\dot{v}(x) + v(x)\dot{u}(x)$

4.2.6 Quotient of Two Functions.

If we have the function $y = f(x) = \frac{u(x)}{v(x)}$, where $u(x)$, and $v(x)$ are differentiable functions at x , also $v(x) \neq 0$. Then the derivative $\dot{y} = \frac{[v(x)\dot{u}(x) - u(x)\dot{v}(x)]}{[v(x)]^2}$

Example 4.1 Use the definition of the derivative to find $\dot{y}$ to each of the following functions.

1. $y = \sqrt{3x - 4}$

$$\begin{aligned}\dot{y} &= \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{3(x+\Delta x) - 4} - \sqrt{3x - 4}}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{3(x+\Delta x) - 4} - \sqrt{3x - 4}}{\Delta x} \cdot \frac{\sqrt{3(x+\Delta x) - 4} + \sqrt{3x - 4}}{\sqrt{3(x+\Delta x) - 4} + \sqrt{3x - 4}} \\ &= \lim_{\Delta x \rightarrow 0} \frac{3x + 3\Delta x - 4 - 3x + 4}{\Delta x (\sqrt{3(x+\Delta x) - 4} + \sqrt{3x - 4})} \\ &= \lim_{\Delta x \rightarrow 0} \frac{3}{\sqrt{3(x + \Delta x) - 4} + \sqrt{3x - 4}} \\ &= \frac{3}{2\sqrt{3x - 4}}\end{aligned}$$



2. $y = \sqrt{x}$

$$\begin{aligned}\dot{y} &= \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{x+\Delta x} - \sqrt{x}}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{x+\Delta x} - \sqrt{x}}{\Delta x} \frac{\sqrt{x+\Delta x} + \sqrt{x}}{\sqrt{x+\Delta x} + \sqrt{x}} = \lim_{\Delta x \rightarrow 0} \frac{x+\Delta x - x}{\Delta x(\sqrt{x+\Delta x} + \sqrt{x})} \\&= \lim_{\Delta x \rightarrow 0} \frac{1}{\sqrt{x+\Delta x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}\end{aligned}$$

3. $y = \frac{5}{x}$

$$\begin{aligned}\dot{y} &= \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{\frac{5}{x+\Delta x} - \frac{5}{x}}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{\frac{5x - 5(x+\Delta x)}{(x+\Delta x)x}}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\frac{5x - 5x - 5\Delta x}{(x+\Delta x)x}}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{-5}{(x+\Delta x)x} = \frac{-5}{x^2}\end{aligned}$$

4. $y = 2x^2 - 16x + 35$

$$\begin{aligned}\dot{y} &= \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{2(x+\Delta x)^2 - 16(x+\Delta x) + 35 - (2x^2 - 16x + 35)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{2(x^2 + 2x\Delta x + \Delta x^2) - 16x - 16\Delta x + 35 - 2x^2 + 16x - 35}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{2x^2 + 4x\Delta x + 2\Delta x^2 - 16\Delta x - 2x^2}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{4x\Delta x + 2\Delta x^2 - 16\Delta x}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{\Delta x(4x + 2\Delta x - 16)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} (4x + 2\Delta x - 16) = 4x - 16\end{aligned}$$



5. $y = \sin(x)$

$$\dot{y} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\sin(x+\Delta x) - \sin(x)}{\Delta x}$$

Using angles sum identity, we get

$$\begin{aligned} &= \lim_{\Delta x \rightarrow 0} \frac{\sin(x) \cos(\Delta x) + \cos(x) \sin(\Delta x) - \sin(x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\cos(x) \sin(\Delta x) - \sin(x) + \sin(x) \cos(\Delta x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\cos(x) \sin(\Delta x) - \sin(x) [1 - \cos(\Delta x)]}{\Delta x} \\ &= \cos(x) \lim_{\Delta x \rightarrow 0} \frac{\sin(\Delta x)}{\Delta x} - \sin(x) \lim_{\Delta x \rightarrow 0} \frac{1 - \cos(\Delta x)}{\Delta x} \end{aligned}$$

Using the two special limits, $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$, and $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$,

$$= \cos(x)(1) - \sin(x)(0) = \cos(x)$$

6. $y = \cos(3x)$

$$\dot{y} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\cos[3(x+\Delta x)] - \cos(3x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\cos(3x+3\Delta x) - \cos(3x)}{\Delta x}$$

Using angles sum identity, we get,

$$\begin{aligned} &= \lim_{\Delta x \rightarrow 0} \frac{\cos(3x) \cos(3\Delta x) - \sin(3x) \sin(3\Delta x) - \cos(3x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\cos(3x) [\cos(3\Delta x) - 1] - \sin(3x) \sin(3\Delta x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\cos(3x) [\cos(3\Delta x) - 1] - \sin(3x) \sin(3\Delta x)}{\Delta x} \end{aligned}$$

Using the two special limits, $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$, and $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$,

$$\begin{aligned} &= \lim_{\Delta x \rightarrow 0} \left[\cos(3x) \frac{\cos(3\Delta x) - 1}{\Delta x} - \sin(3x) \frac{\sin(3\Delta x)}{\Delta x} \right] \\ &= \lim_{\Delta x \rightarrow 0} \left[-3\cos(3x) \frac{1 - \cos(3\Delta x)}{3\Delta x} - 3\sin(3x) \frac{\sin(3\Delta x)}{3\Delta x} \right] \end{aligned}$$

as $\Delta x \rightarrow 0$, then $3\Delta x \rightarrow 0$,

$$\begin{aligned} &= -3\cos(3x) \lim_{3\Delta x \rightarrow 0} \frac{1 - \cos(3\Delta x)}{3\Delta x} - 3\sin(3x) \lim_{3\Delta x \rightarrow 0} \frac{\sin(3\Delta x)}{3\Delta x} \\ &= 3\cos(3x)(0) - 3\sin(3x)(1) = -3\sin(3x) \end{aligned}$$



4.3 Differentiation Formulas.

In this section a table lists different functions, and their corresponding first derivative, which is a required knowledge for the coming differentiation methods.

Function	First Derivative
1. $f[u(x)]$	$f'[u(x)] \cdot u'(x)$
2. x^n	$n \cdot x^{n-1}$
3. $[u(x)]^n$	$n \cdot [u(x)]^{n-1} \cdot u'(x)$
4. $[f[u(x)]]^n$	$n \cdot [f[u(x)]]^{n-1} \cdot f'[u(x)] \cdot u'(x)$

Exponential Functions

5. e^x	e^x
6. $e^{u(x)}$	$e^{u(x)} \cdot u'(x)$
7. a^x	$a^x \cdot \ln(a)$
8. $a^{u(x)}$	$a^{u(x)} \cdot \ln(a) \cdot u'(x)$

Logarithmic Functions

9. $\ln(x)$	$\frac{1}{x}$
10. $\ln[u(x)]$	$\frac{1}{u(x)} \cdot u'(x)$
11. $\log_a(x)$	$\frac{1}{x \cdot \ln(a)}$
12. $\log_a[u(x)]$	$\frac{1}{u(x) \cdot \ln(a)} \cdot u'(x)$

Trigonometric Functions

13. $\sin(x)$	$\cos(x)$
14. $\cos(x)$	$-\sin(x)$
15. $\tan(x)$	$\sec^2(x)$
16. $\cot(x)$	$-\csc^2(x)$
17. $\sec(x)$	$\sec(x) \tan(x)$
18. $\csc(x)$	$-\csc(x) \cot(x)$



Inverse Trigonometric Functions

- | | |
|--------------------|----------------------------|
| 19. $\sin^{-1}(x)$ | $\frac{1}{\sqrt{1-x^2}}$ |
| 20. $\cos^{-1}(x)$ | $-\frac{1}{\sqrt{1-x^2}}$ |
| 21. $\tan^{-1}(x)$ | $\frac{1}{1+x^2}$ |
| 22. $\sec^{-1}(x)$ | $\frac{1}{x\sqrt{x^2-1}}$ |
| 23. $\csc^{-1}(x)$ | $-\frac{1}{x\sqrt{x^2-1}}$ |
| 24. $\cot^{-1}(x)$ | $-\frac{1}{1+x^2}$ |

Hyperbolic Functions

- | | |
|------------------------------|-----------------------------------|
| 25. $\sinh(x)$ | $\cosh(x)$ |
| 26. $\cosh(x)$ | $\sinh(x)$ |
| 27. $\tanh(x)$ | $\operatorname{sech}^2(x)$ |
| 28. $\coth(x)$ | $-\operatorname{csch}^2(x)$ |
| 29. $\operatorname{sech}(x)$ | $-\operatorname{sech}(x)\tanh(x)$ |
| 30. $\operatorname{csch}(x)$ | $-\operatorname{csch}(x)\coth(x)$ |

Inverse Hyperbolic Functions

- | | |
|-----------------------------------|---|
| 31. $\sinh^{-1}(x)$ | $\frac{1}{\sqrt{1+x^2}}$ |
| 32. $\cosh^{-1}(x)$ | $\frac{1}{\sqrt{x^2-1}}$ |
| 33. $\tanh^{-1}(x)$ | $\frac{1}{1-x^2}$ |
| 34. $\operatorname{sech}^{-1}(x)$ | $-\frac{1}{x(x+1)\sqrt{\frac{1-x}{1+x}}}$ |
| 35. $\operatorname{csch}^{-1}(x)$ | $-\frac{1}{x^2\sqrt{1+\frac{1}{x^2}}}$ |
| 36. $\coth^{-1}(x)$ | $\frac{1}{1-x^2}$ |



Example 4.2 find the derivative of the given functions.

1. $y = (4x - 5)^3$

$$\begin{aligned}\dot{y} &= 3(4x - 5)^2 \cdot 4 \\ &= 12(4x - 5)^2\end{aligned}$$

2. $y = e^{3-x}$

$$\begin{aligned}\dot{y} &= e^{3-x} \times -1 \\ &= -e^{3-x}\end{aligned}$$

3. $y = \ln(3 - 4 \cos(x))$

$$\begin{aligned}\dot{y} &= \frac{1}{3-4 \cos(x)} [0 - 4(-\sin(x))] \\ &= \frac{4 \sin(x)}{3-4 \cos(x)}\end{aligned}$$

4. $y = x \sinh(x) - \cosh(x)$

$$\begin{aligned}\dot{y} &= x \cosh(x) + \sinh(x) - \sinh(x) \\ &= x \cosh(x)\end{aligned}$$

5. $y = \tan^{-1}\left(\frac{1}{x}\right)$

$$\begin{aligned}\dot{y} &= \frac{1}{1 + \left(\frac{1}{x}\right)^2} \cdot \frac{-1}{x^2} = \frac{1}{1 + \frac{1}{x^2}} \cdot \frac{-1}{x^2} \\ &= \frac{1}{\frac{x^2 + 1}{x^2}} \cdot \frac{-1}{x^2} = \frac{x^2}{x^2 + 1} \cdot \frac{-1}{x^2} \\ &= \frac{-1}{x^2 + 1}\end{aligned}$$



Example 4.3 show that the derivative of $y = \sinh(x)$ is $\dot{y} = \cosh(x)$.

$$\begin{aligned}y &= \sinh(x) = \frac{e^x - e^{-x}}{2} \\&= \frac{1}{2}(e^x - e^{-x}) \\ \dot{y} &= \frac{1}{2}(e^x \times 1 - e^{-x} \times -1) \\&= \frac{1}{2}(e^x + e^{-x}) = \cosh(x)\end{aligned}$$

H.W.15 find the derivative $\dot{y}$ of the given functions.

$$y = e^{\sin(2x)}$$

$$y = \ln[\cos(3x)]$$

$$y = \cos^3(3x)$$

$$y = \log(2x - 1)$$

$$y = e^{(2x)} \ln(5x)$$

$$y = x^2 \tan(x)$$

$$y = x^2 \ln[\sinh(x)]$$

$$y = \frac{\sin(3x)}{x+1}$$

$$y = \frac{\ln(x)}{e^{(2x)}}$$



4.4 Implicit differentiation.

To this point we've done quite a few derivatives, but they have all been derivatives of functions of the form $y = f(x)$. Unfortunately, not all the functions that we're going to look at will fall into this form. It is impossible to algebraically express y explicitly as a function of x , and therefore one cannot find $\frac{dy}{dx}$ by explicit differentiation. Instead, one can totally differentiate $z(x, y) = 0$ with respect to x and y and then solve the resulting linear equation for $\frac{dy}{dx}$ to explicitly get the derivative in terms of x and y .

Example 4.4 Differentiate each of the following.

1. $x^2 + y^2 = 25$

$$2x + 2y\dot{y} = 0 \rightarrow 2y\dot{y} = -2x$$

$$\dot{y} = -\frac{2x}{2y} = -\frac{x}{y}$$

2. $x^2 \tan(y) + y^{10} \sec(x) = 2x$

$$x^2 \sec^2(y) \dot{y} + \tan(y) 2x + y^{10} \sec(x) \tan(x) + \sec(x) 10y^9 \dot{y} = 2$$

$$x^2 \sec^2(y) \dot{y} + 10 \sec(x) y^9 \dot{y} = 2 - 2x \tan(y) - \sec(x) \tan(x) y^{10}$$

$$[x^2 \sec^2(y) + 10 \sec(x) y^9] \dot{y} = 2 - 2x \tan(y) - \sec(x) \tan(x) y^{10}$$

$$\dot{y} = \frac{2 - 2x \tan(y) - \sec(x) \tan(x) y^{10}}{x^2 \sec^2(y) + 10 \sec(x) y^9}$$

3. $e^{2x+3y} = x^2 - \ln(xy^3)$

$$e^{2x+3y} (2 + 3\dot{y}) = 2x - \frac{1}{xy^3} (x3y^2\dot{y} + y^3)$$

$$2e^{2x+3y} + 3e^{2x+3y}\dot{y} = 2x - \left(\frac{3xy^2\dot{y}}{xy^3} + \frac{y^3}{xy^3}\right)$$

$$2e^{2x+3y} + 3e^{2x+3y}\dot{y} = 2x - \frac{3}{y}\dot{y} - \frac{1}{x}$$

$$3e^{2x+3y}\dot{y} + \frac{3}{y}\dot{y} = 2x - \frac{1}{x} - 2e^{2x+3y}$$

$$\left(3e^{2x+3y} + \frac{3}{y}\right)\dot{y} = 2x - \frac{1}{x} - 2e^{2x+3y}$$

$$\dot{y} = \frac{2x - \frac{1}{x} - 2e^{2x+3y}}{3e^{2x+3y} + \frac{3}{y}}$$



H.W.16 Find $\dot{y}$ for each of the following.

1. $x^2 + y^3 + 3xy^2 = 8$.
2. $x^3y^5 + 3x = 8y^3 + 1$.
3. $x^3y^6 + e^{1-x} - \cos(5y) = y^2$.

4.5 Logarithmic Differentiation.

In calculus, logarithmic differentiation is a method used to differentiate functions after taking logarithms first. This approach allows calculating derivatives of power, product, rational, and some irrational functions in situations where it is easier to differentiate the logarithm of a function than to differentiate the function itself.

Consider the function $y = f(x)$ is to be differentiated. Taking the natural logarithm first yields,

$$\ln y = \ln f(x)$$

now, we differentiate using implicit differentiation,

$$\frac{1}{y} \frac{dy}{dx} = \frac{d}{dx} [\ln f(x)]$$

Hence the derivative is

$$\frac{dy}{dx} = y \frac{d}{dx} [\ln f(x)] = f(x) \frac{d}{dx} [\ln f(x)]$$

Example 4.5 Differentiate the following functions.

1. $y = \frac{x^3 \sin(x)}{\cos(2x)}$

Solution.

$$\begin{aligned} \ln(y) &= \ln \left[\frac{x^3 \sin(x)}{\cos(2x)} \right] = \ln[(x^3) \sin(x)] - \ln[\cos(2x)] \\ &= \ln(x^3) + \ln[\sin(x)] - \ln[\cos(2x)] \end{aligned}$$

$$\frac{1}{y} \dot{y} = \frac{1}{x^3} 3x^2 + \frac{1}{\sin(x)} \cos(x) - \frac{1}{\cos(2x)} (-\sin(2x) 2)$$

$$\frac{1}{y} \dot{y} = \frac{3}{x} + \cot(x) + 2 \tan(2x)$$

$$\begin{aligned} \dot{y} &= y \left[\frac{3}{x} + \cot(x) + 2 \tan(2x) \right] \\ &= \frac{x^3 \sin(x)}{\cos(2x)} \left[\frac{3}{x} + \cot(x) + 2 \tan(2x) \right] \end{aligned}$$



$$2. y = \frac{x^5}{(1-10x)\sqrt{x^2+2}}$$

Solution.

$$\begin{aligned}\ln(y) &= \ln \left[\frac{x^5}{(1-10x)\sqrt{x^2+2}} \right] \\ &= \ln(x^5) - \ln[(1-10x)\sqrt{x^2+2}] \\ &= \ln(x^5) - [\ln(1-10x) + \ln(\sqrt{x^2+2})] \\ &= \ln(x^5) - \ln(1-10x) - \ln(\sqrt{x^2+2}) \\ \frac{1}{y} \dot{y} &= \frac{1}{x^5} 5x^4 - \frac{1}{1-10x} (-10) - \frac{1}{\sqrt{x^2+2}} \frac{1}{2} \frac{1}{\sqrt{x^2+2}} 2x \\ \frac{1}{y} \dot{y} &= \frac{5}{x} + \frac{10}{1-10x} - \frac{x}{x^2+2} \\ \dot{y} &= y \left(\frac{5}{x} + \frac{10}{1-10x} - \frac{x}{x^2+2} \right) \\ &= \frac{x^5}{(1-10x)\sqrt{x^2+2}} \left(\frac{5}{x} + \frac{10}{1-10x} - \frac{x}{x^2+2} \right)\end{aligned}$$

H.W.17 Differentiate the following functions.

1. $y = x^x$
2. $y = x^4 e^{3x} \tan(x)$
3. $y = x \ln^2(xe^{3x}) \tan(x) \tanh(x)$
4. $y = (1-3x)^{\cos(x)}$
5. $y = (5-3x^2)^7 \sqrt{6x^2+8x-12}$
6. $y = \frac{\sqrt{5x+8} \sqrt[3]{1-9\cos(4x)}}{\sqrt{x^2+10x}}$



4.6 Chain Rule.

We've taken a lot of derivatives over the course of our differentiation lectures. However, if you look back, they have all been direct functions. These are all simple functions in that wherever the variable appears it is by itself.

What about functions like $y = \cos(2t)$ and $x = \sin(t)$, and it is required from you to find $\frac{dy}{dx}$. In mathematical analysis, the chain rule is a derivation rule that allows to calculate the derivative of the function composed of two or more derivable functions. In this case,

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}$$

Furthermore, $\frac{d^2y}{dx^2}$ then can be found by using the following rule,

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx}$$

Example 4.6 find $\frac{dy}{dx}$, and $\frac{d^2y}{dx^2}$ to each of the following functions.

1. $y = \cos(2t)$, $x = \sin(t)$

Solution.

$$\begin{aligned} \frac{dy}{dt} &= -\sin(2t) \cdot 2 \\ &= -2 \sin(2t) \end{aligned}$$

$$\frac{dx}{dt} = \cos(t)$$

$$\begin{aligned} \frac{dt}{dx} &= \frac{1}{\frac{dx}{dt}} \\ &= \frac{1}{\cos(t)} \end{aligned}$$



$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}$$

$$= [-2 \sin(2t)] \left[\frac{1}{\cos(t)} \right]$$

$$= \frac{-2 \sin(2t)}{\cos(t)}$$

$$= \frac{-2(2 \sin(t) \cos(t))}{\cos(t)}$$

$$= -4 \sin(t)$$

$$\frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{d}{dt} [-4 \sin(t)]$$

$$= -4 \cos(t)$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx}$$

$$= [-4 \cos(t)] \left[\frac{1}{\cos(t)} \right] = -4$$

$$2. \ y = \frac{3+2t}{1+t}, \ x = \frac{2-3t}{1+t}$$

Solution.

$$\frac{dy}{dt} = \frac{(1+t)(2) - (3+2t)(1)}{(1+t)^2}$$

$$= \frac{2 + 2t - 3 - 2t}{(1+t)^2}$$

$$= \frac{-1}{(1+t)^2}$$

$$\frac{dx}{dt} = \frac{(1+t)(-3) - (2-3t)(1)}{(1+t)^2}$$

$$= \frac{-3 - 3t - 2 + 3t}{(1+t)^2}$$

$$= \frac{-5}{(1+t)^2}$$



$$\frac{dt}{dx} = \frac{1}{\frac{dx}{dt}} = \frac{1}{\frac{-5}{(1+t)^2}}$$
$$= \frac{(1+t)^2}{-5}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}$$
$$= \left[\frac{-1}{(1+t)^2} \right] \left[\frac{(1+t)^2}{-5} \right] = \frac{1}{5}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx}$$
$$= \frac{d}{dt} \left(\frac{1}{5} \right) \left[\frac{(1+t)^2}{-5} \right] = 0 \times \frac{(1+t)^2}{-5} = 0$$

Example 4.7 Given the function $y = \frac{x}{x^2+1}$, find $\frac{d\sqrt{y}}{d\sqrt{x+1}}$

Solution.

Let $u = \sqrt{y}$, and $v = \sqrt{x+1}$, then to find $\frac{du}{dv}$ we use the chain rule to have,

$$\frac{du}{dv} = \frac{du}{dy} \frac{dy}{dx} \frac{dx}{dv}$$

$$\frac{du}{dy} = \frac{1}{2\sqrt{y}}$$

$$\frac{dy}{dx} = \frac{(x^2+1)(1) - (x)(2x)}{(x^2+1)^2}$$
$$= \frac{x^2+1-2x^2}{(x^2+1)^2}$$
$$= \frac{1-x^2}{(x^2+1)^2}$$

$$\frac{dv}{dx} = \frac{1}{2\sqrt{x+1}} \times (1)$$
$$= \frac{1}{2\sqrt{x+1}}$$



$$\frac{dx}{dv} = \frac{1}{\frac{dv}{dx}} = \frac{1}{\frac{1}{2\sqrt{x+1}}} \\ = 2\sqrt{x+1}$$

$$\frac{d\sqrt{y}}{d\sqrt{x+1}} = \frac{du}{dv} \\ = \frac{du}{dy} \frac{dy}{dx} \frac{dx}{dv} \\ = \left[\frac{1}{2\sqrt{y}} \right] \left[\frac{1-x^2}{(x^2+1)^2} \right] [2\sqrt{x+1}] \\ = \left[\frac{1}{2 \times \sqrt{\frac{x}{x^2+1}}} \right] \left[\frac{1-x^2}{(x^2+1)^2} \right] [2\sqrt{x+1}] \\ = \frac{(1-x^2)(2\sqrt{x+1})}{\left(2 \times \sqrt{\frac{x}{x^2+1}} \right) (x^2+1)^2}$$

H.W.18 find $\frac{dy}{dx}$, and $\frac{d^2y}{dx^2}$ of each of the following pair of functions.

- a) $y = 3 \sin(\theta) - \sin^3(\theta)$, $x = \cos^3(\theta)$.
- b) $y = \alpha[\sin(\theta) - \theta \cos(\theta)]$, $x = \alpha[\cos(\theta) + \theta \sin(\theta)]$.
- c) $y = \tan(\theta) - \theta$, $x = \ln \left[\tan \left(\frac{\theta}{2} \right) \right]$.
- d) $y = 3[\theta - \sin(\theta)]$, $x = 3[1 - \cos(\theta)]$.

H.W.19 Given the function $y = x^3 - 2x^2 + 3x - 1$, find $\frac{dy^2}{dx^2}$.